

UDP Lab

The image shows a Wireshark capture of network traffic. The top pane displays a list of 20 UDP packets. The bottom pane shows the details of the selected packet (No. 1), which is a Multicast Domain Name System (MDNS) query. The packet structure is as follows:

- Internet Protocol Version 4, Src: 10.20.10.163, Dst: 224.0.0.251
- User Datagram Protocol, Src Port: 5353, Dst Port: 5353
 - Source Port: 5353
 - Destination Port: 5353
 - Length: 69
 - Checksum: 0x339d [unverified]
 - [Checksum Status: Unverified]
 - [Stream index: 0]
 - [Stream Packet Number: 1]
 - [Timestamps]
 - UDP payload (61 bytes)
 - Multicast Domain Name System (query)

The packet bytes pane shows the raw data of the packet, starting with the Ethernet II header and followed by the IP and UDP headers and payload.

- the 4 UDP header fields are:
 - Source Port
 - Destination Port
 - Length
 - Checksum
- Each is 2 bytes. Total header = 8 bytes
- Length field value = 69 bytes
 - UDP header = 8 bytes
 - UDP payload = 61 bytes ← Wireshark even confirms "UDP payload (61 bytes)"
 - 69 = 8 + 61 – So Length = header + payload combined**
- Max length field value = 65,535 - 8 byte header = 65,527 bytes max payload
- Largest possible source port = 65,535

UDP Lab

Wireshark capture of UDP traffic. The packet list shows a series of MDNS queries from 10.20.10.163 to 224.0.0.251. The packet details pane shows the structure of a UDP packet, including Ethernet II, Internet Protocol Version 4, and User Datagram Protocol. The packet bytes pane shows the raw hex and ASCII data.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	10.20.10.163	224.0.0.251	MDNS	103	Standard query 0x0000 TXT Asritha's Laptop_companion-link_tcp...
2	0.000001000	10.20.9.203	224.0.0.251	MDNS	109	Standard query 0x0000 TXT 수민의 MacBook Air_companion-link_tc...
3	0.000001500	10.20.10.131	224.0.0.251	MDNS	103	Standard query 0x0000 TXT Asritha's Laptop_companion-link_tcp...
4	0.000002600	10.20.11.92	224.0.0.251	MDNS	1494	Standard query 0x0000 PTR _companion-link_tcp.local, "QM" ques...
5	0.000003600	10.20.10.180	224.0.0.251	MDNS	278	Standard query 0x0000 TXT Idil's MacBook Air_airplay_tcp.local...
6	0.001368500	10.20.9.206	224.0.0.251	MDNS	139	Standard query 0x0000 TXT Anya's MacBook Air_airplay_tcp.local...
7	0.003581600	10.20.11.92	224.0.0.251	MDNS	1427	Standard query 0x0000 PTR Kala's MacBook Air_companion-link_tcp...
8	0.003583000	10.20.9.107	224.0.0.251	MDNS	103	Standard query 0x0000 TXT Asritha's Laptop_companion-link_tcp...
9	0.003583700	10.20.10.45	224.0.0.251	MDNS	1475	Standard query 0x0000 PTR _airplay_tcp.local, "QM" question PT...
10	0.008446800	10.20.10.45	224.0.0.251	MDNS	999	Standard query 0x0000 PTR Brandon Macbook_airplay_tcp.local...
11	0.008448100	10.20.10.121	224.0.0.251	MDNS	1494	Standard query 0x0000 PTR _airplay_tcp.local, "QM" question PT...
12	0.008453000	10.20.10.121	224.0.0.251	MDNS	734	Standard query 0x0000 PTR Delaney's MacBook Air_airplay_tcp.l...
13	0.008454200	10.20.11.126	224.0.0.251	MDNS	132	Standard query 0x0000 TXT 수민의 MacBook Air_companion-link_tc...
14	0.008454800	10.20.10.45	224.0.0.251	MDNS	132	Standard query 0x0000 TXT Asritha's Laptop_companion-link_tcp...
15	0.008804200	10.20.11.226	224.0.0.251	MDNS	132	Standard query 0x0000 TXT Asritha's Laptop_companion-link_tcp...
16	0.008804300	10.20.11.157	224.0.0.251	MDNS	107	Standard query 0x0000 TXT Meba's MacBook Air_companion-link_tcp...
17	0.0088043800	10.20.10.197	224.0.0.251	MDNS	103	Standard query 0x0000 TXT Sophia is goats_companion-link_tcp...
18	0.0088044400	10.20.10.209	224.0.0.251	MDNS	140	Standard query 0x0000 TXT Sophia is goats_companion-link_tcp...
19	0.0088044900	10.20.9.152	224.0.0.251	MDNS	137	Standard query 0x0000 TXT Meadow's MacBook Air_companion-link...
20	0.0088045700	10.20.9.199	239.255.255.250	SSDP	167	M-SEARCH * HTTP/1.1

Ethernet II, Src: Apple 97:d7:2f (98:01:a7:97:d7:2f), Dst: IPv4mcast_fb (01:00:5e:00:00:00)

Internet Protocol Version 4, Src: 10.20.10.163, Dst: 224.0.0.251

0100 = Version: 4

.... 0101 = Header Length: 20 bytes (5)

Differentiated Services Field: 0xc0 (DSCP: CS6, ECN: Not-ECT)

Total Length: 89

Identification: 0x9d9f (40447)

0000 = Flags: 0x0

... 0000 0000 0000 = Fragment Offset: 0

Time to Live: 255

Protocol: UDP (17)

Header Checksum: 0x2722 [validation disabled]

Source Port (udp.srcport), 2 bytes

Packets: 2011 · Displayed: 1890 (94.0%) · Dropped: 0 (0.0%) · Profile: Default

6. Decimal: 17 Hexadecimal: 0x11

Wireshark capture of DNS traffic. The packet list shows a series of DNS queries and responses from 10.20.11.119 to 192.168.45.133. The packet details pane shows the structure of a DNS packet, including Ethernet II, Internet Protocol Version 4, and User Datagram Protocol. The packet bytes pane shows the raw hex and ASCII data.

No.	Time	Source	Destination	Protocol	Length	Info
1118	1.578702100	10.20.11.119	192.168.45.133	DNS	70	Standard query 0x53ce A google.com
1132	1.835950700	10.20.11.119	192.168.45.133	DNS	70	Standard query 0x53ce A google.com
1144	2.842037300	10.20.11.119	192.168.45.133	DNS	70	Standard query 0x53ce A google.com
1146	4.848181900	10.20.11.119	192.168.45.133	DNS	70	Standard query 0x53ce A google.com
1147	4.848362100	10.20.11.119	192.168.45.133	DNS	70	Standard query 0x53ce A google.com
2423	8.817574900	10.20.11.119	192.168.45.133	DNS	83	Standard query 0x9375 HTTPS bl.nel.goog
2425	8.817729400	10.20.11.119	192.168.45.133	DNS	83	Standard query 0x66e9 A bl.nel.goog
2467	8.844989500	192.168.45.133	10.20.11.119	DNS	101	Standard query response 0x66e9 A bl.nel.goog A 142.250.137.94
2468	8.844989300	192.168.45.133	10.20.11.119	DNS	145	Standard query response 0x9375 HTTPS bl.nel.goog SOA ns1.google.com
2477	8.853666400	10.20.11.119	192.168.45.133	DNS	70	Standard query 0x53ce A google.com
2478	8.853707100	10.20.11.119	192.168.45.133	DNS	70	Standard query 0x53ce A google.com
2487	8.859794900	192.168.45.133	10.20.11.119	DNS	86	Standard query response 0x53ce A google.com A 142.250.191.14
2488	8.859815100	192.168.45.133	10.20.11.119	DNS	166	Standard query response 0x53ce A google.com A 142.250.137.100 A 142.2...
2548	8.982655700	10.20.11.119	192.168.45.133	DNS	95	Standard query 0x62cd A optimizationguide-pa.googleapis.com
2549	8.982883900	10.20.11.119	192.168.45.133	DNS	95	Standard query 0x9e9f HTTPS optimizationguide-pa.googleapis.com
2627	9.178150600	192.168.45.133	10.20.11.119	DNS	159	Standard query response 0x62cd A optimizationguide-pa.googleapis.com ...
2630	9.188637200	192.168.45.133	10.20.11.119	DNS	152	Standard query response 0x9e9f HTTPS optimizationguide-pa.googleapis...

Frame 1118: Packet, 70 bytes on wire (560 bits), 70 bytes captured (560 bits) on interface 0

Ethernet II, Src: CloudNetwork_8b:f6:df (48:45:e6:8b:f6:df), Dst: Cisco_2e:d0:10 (12:00:00:00:00:00)

Internet Protocol Version 4, Src: 10.20.11.119, Dst: 192.168.45.133

User Datagram Protocol, Src Port: 63515, Dst Port: 53

Source Port: 63515

Destination Port: 53

Length: 36

Checksum: 0x03ee [unverified]

[Checksum Status: Unverified]

[Stream index: 247]

[Stream Packet Number: 1]

[Timestamps]

wireshark_Wi-FiUSJ3Q3.pcapng

Packets: 3528 · Displayed: 17 (0.5%) · Dropped: 0 (0.0%) · Profile: Default

7. Query (your host sends):

- Source Port: 63515 (random ephemeral port your OS picked)
- Destination Port: 53 (DNS server)

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Response (reply comes back):

- a. Source Port: 53 (DNS server)
- b. Destination Port: 63515 (back to you)